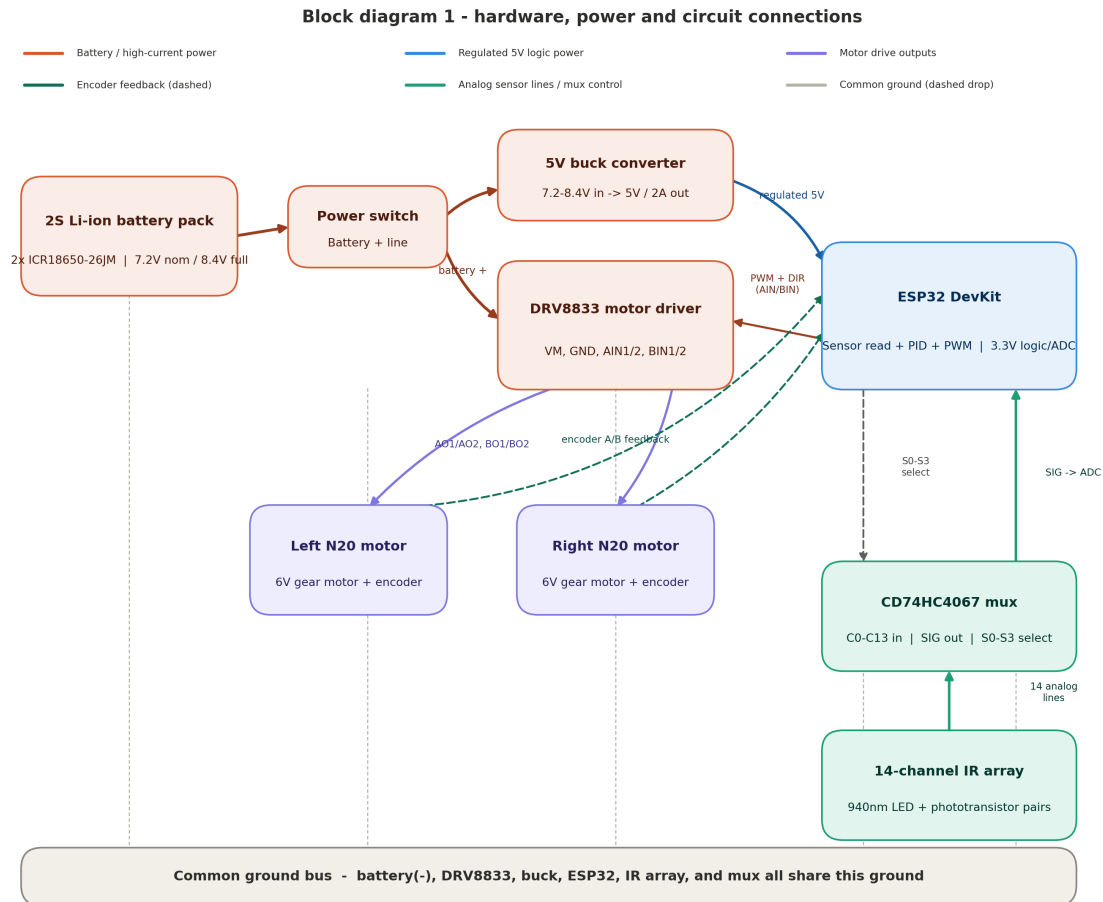


Fast line-following robot

Final block diagrams | ESP32 · DRV8833 · 2x N20 encoder motors · 14-channel analog IR array · CD74HC4067 multiplexer

1. Hardware, power and circuit connections

Power flows from the 2S Li-ion pack through the main switch to two regulators: the DRV8833 takes raw battery voltage for the motors, and the buck converter steps it down to a regulated 5V rail for the ESP32 and sensor electronics. The ESP32 closes the loop by reading the IR array through the multiplexer and driving the motors through the DRV8833, using encoder feedback to track wheel speed. Every module shares one common ground.



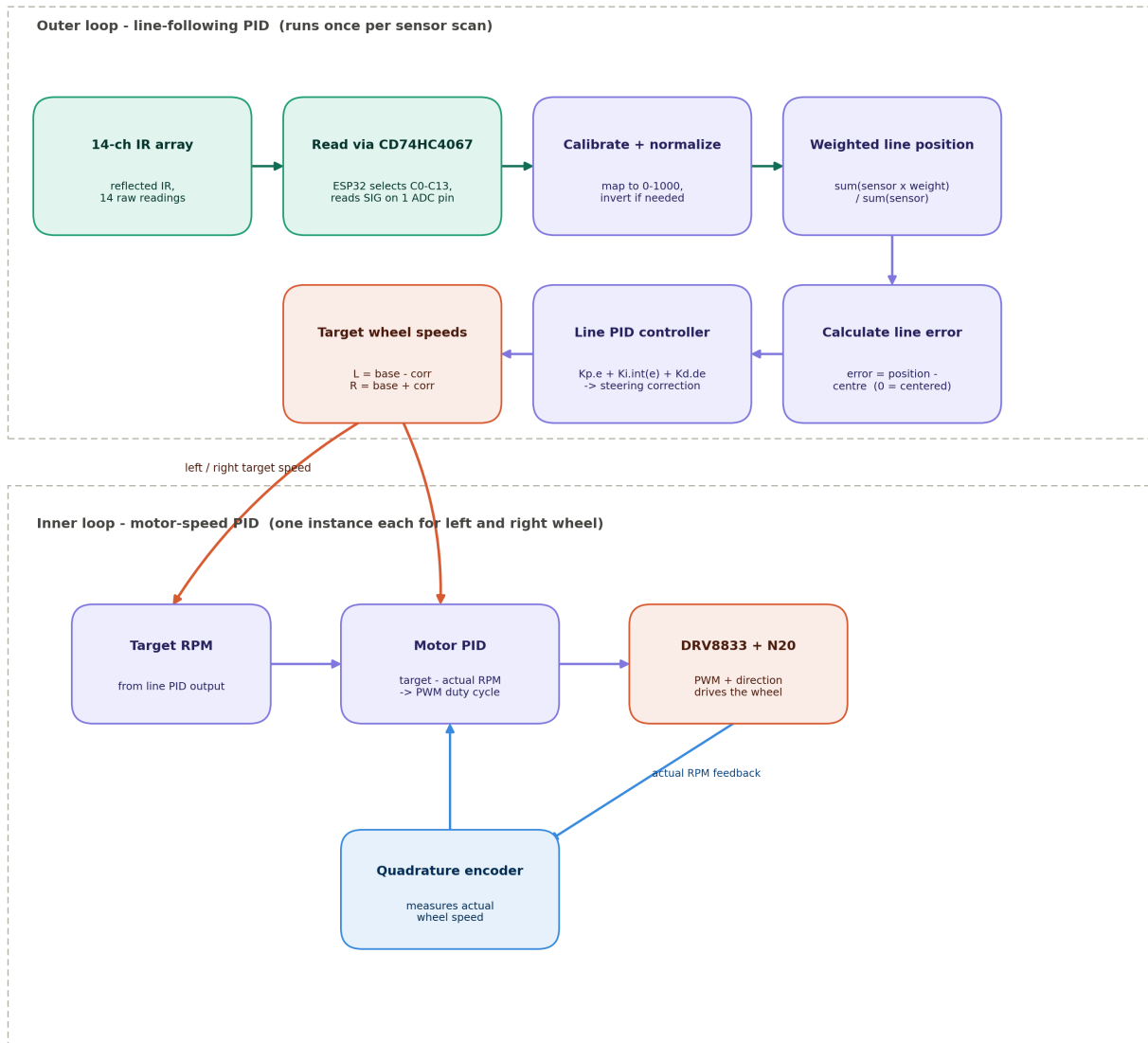
Critical wiring notes

- The battery's 8.4V never connects to the ESP32 directly - only the DRV8833 motor supply (VM) sees raw battery voltage.
- The ESP32 is powered exclusively from the regulated 5V buck converter output (5V / VIN pin).
- All six grounds - battery negative, DRV8833, buck converter, ESP32, IR array, and multiplexer - must tie to one common ground.
- DRV8833 STBY must be held HIGH (tie to 3.3V or an ESP32 GPIO driven HIGH), or the motor outputs stay disabled.
- Add a 470-1000 uF electrolytic capacitor across DRV8833 VM-GND, close to the driver, plus 100 nF ceramic capacitors near the ESP32 and multiplexer supply pins to suppress motor-switching noise.

2. Two-level control loop

The firmware runs two nested PID loops. The outer loop reads all 14 IR sensors once per scan, computes a weighted line position, and turns the resulting error into a steering correction and a pair of target wheel speeds. The inner loop runs once per wheel: it compares the target RPM against the actual RPM measured by the quadrature encoder and adjusts PWM duty cycle to close the gap.

Block diagram 2 - two-level control loop



How steering happens

- Line left of center → correction is negative → left target speed decreases, right target speed increases → robot turns left.
- Line right of center → correction is positive → left target speed increases, right target speed decreases → robot turns right.
- For sharp turns the firmware may reverse one motor - no mechanical differential or steering servo is required.

3. Final component reference

This table matches the two diagrams above. Items marked “optional” strengthen a physical build (power filtering, fusing) but can be omitted only for idealized simulation.

Subsystem	Component / circuit	Qty	Key specification / connection
Controller	ESP32 DevKit	1	ESP32-WROOM-32; 3.3V GPIO/ADC; powered from regulated 5V VIN/5V pin
Sensors	940 nm IR LED	14	One per sensing position, each with a 220 Ω series resistor
Sensors	Phototransistor	14	One per IR LED, each with a 10 k Ω load/pull-down resistor; analog output
Sensors	CD74HC4067	1	16:1 analog multiplexer; C0-C13 from sensors, SIG to ESP32 ADC, S0-S3 from ESP32
Motors	N20 DC gear motor + quadrature encoder	2	6V class; left and right wheels driven independently
Motor driver	DRV8833 breakout board	1	VM from battery; AIN1/2, BIN1/2 from ESP32; STBY held HIGH; AO/BO to motors
Battery	Samsung ICR18650-26JM	2	Identical cells in series (2S): 7.2V nominal, 8.4V full
Battery safety	2S BMS + 8.4V Li-ion charger	1 each	BMS rated at least 8-10 A; charger designed for 2S Li-ion
Power	5V buck converter	1	Input 7.2-8.4V; output 5V; rated $\geq$ 2A for ESP32 and sensor electronics
Power	Switch, fuse, wiring, pack holder	1 set	Switch in battery positive line; optional 3A fuse; common ground everywhere
Filtering (optional)	470-1000 μ F electrolytic capacitor	1	Across DRV8833 VM and GND, close to the driver
Filtering (optional)	100 nF ceramic capacitor	2-3	Near ESP32, multiplexer, and/or driver supply pins
Mechanics	Wheels + low-friction front skid	2 + 1	30-40 mm wheels; skid or caster provides the third contact point

Critical wiring notes

- Do not connect the 8.4V battery rail directly to the ESP32.
- The DRV8833 motor supply (VM) receives raw battery voltage; the ESP32 receives regulated 5V from the buck converter.
- All grounds must be tied together on one common bus.
- DRV8833 STBY must be driven HIGH (3.3V or an ESP32 GPIO set HIGH), or the motor outputs remain disabled.